Bumbum, Mai'Adua, Katsina State, Nigeria

In Mai'Adua, El Niño's peak falls in meningitis season, not heat season. Last season 6 of the LGA's 14 suspected cases died. Put antibiotics, tests, and a ready vehicle at every clinic before December, and settle the household rule that a sick child goes to the clinic the same day.

IMPACT REGION	EXCESS HEAT DEATHS	WINDOW	DOMINANT HAZARD
Not resolved. Mai'Adua LGA (201,178 people, 2006 census; 528 km²), which contains wards Bumbum 'A' and 'B'	Nigeria 31,400 (15,400 in Dec–Feb). Population-scaled to Mai'Adua: ~45, our arithmetic	6-month, Sep 2026 – Feb 2027	Meningitis (Oct–Apr), lean-season child malnutrition (Jun–Sep 2027), cholera with the rains; heat peaks Mar–May

14/14 recommendations checked by verifiers. 2 confirmed, 9 corrected, 2 unverified, 1 contraindicated. Exposure figure: Climate Impact Lab, *451,000 at Risk*, September 2026, pre-peer-review. The ~45 figure is a flat population scaling (201,178 ÷ 140.4 million × 31,400), our arithmetic and not an Impact Lab finding. It also falls mostly in Mai'Adua's cool harmattan season (January mean maximum 29.5°C), so treat it as a national signal, not a local forecast. **The funding target fell 57% in verification**, from the $500,000 first proposed to about $216,000: the first unit cost was per child cured in other states, and two placements could not be verified. **One headline figure was**

EXECUTIVE SUMMARY — AT A GLANCE



13.19°N 8.21°E, Nigeria · Natural Earth

- **The risk.** Meningitis, not heat, is the El Niño-season killer: in 2024/25 Mai'Adua had 14 suspected cases and 6 deaths (43%), and in 2024 a therapeutic-food stock-out preceded over 800 child deaths in MSF's Katsina facilities.
- **First action.** Katsina SPHCDA, the Mai'Adua PHC director, the LGA surveillance officer, and NCDC stock every clinic for meningitis treatment by 30 November 2026, with a referral vehicle and weekly zero-reporting.
- **The money.** ~$120k floor, ~$216k target, ~$592k ceiling. The top placement is therapeutic food and outpatient treatment through Katsina SPHCDA and UNICEF, $80,000–136,000.
- **Do not.** Do not wait for the epidemic threshold to organise a meningitis vaccine request: in 2017 north-west vaccination began about 9 weeks late and only 41% of requested doses arrived.

What to do	**11** actions	p. 2
Who does what	RACI chart	p. 10
Where money does the most	funding	p. 10
Do not do this	**4** items	p. 12
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* Suggested formulation: check with a health worker before sharing.

disputed and resolved: one verifier said Mai'Adua's 6 meningitis deaths belonged to another LGA. We read the NCDC table directly, and the row reads "Katsina Mai'adua 14 6 43".

WHAT THIS PLACE SHOULD DO RANKED

Ranked by expected lives saved per unit of effort before and through the peak. Backing lenses are shown per action; an action several lenses reached independently is the stronger bet. The RACI chart below assigns each one. No Bumbum-specific data exists for any hazard; figures are for Mai'Adua LGA, Katsina State, or Nigeria, and are labelled as such.

1. Stock every Mai'Adua clinic for meningitis treatment by 30 November, with a ready referral vehicle and weekly zero-reporting

PARTIALLY CONFIRMED · EMERGENCY · PUBLIC HEALTH · HISTORIAN · SKEPTIC · FRONTLINE · PHILANTHROPIST

ADAPTATION INVENTORY: NOT CONSULTED

OWNER	BY WHEN	COST	PREVENTS
Katsina SPHCDA with the Mai'Adua PHC director; LGA disease surveillance officer (DSNO); NCDC	Stock by 30 Nov 2026; reporting from now	$40,000–60,000, an unsourced estimate	Deaths from late or missing treatment in the Oct–Apr season

Ceftriaxone, rapid tests, lumbar-puncture kits, and trans-isolate transport media at every primary health centre, including those serving Bumbum, and at Mai'Adua General Hospital. Each facility reports weekly, including zero cases, with escalation within 48 hours. NCDC pre-drafts the request to the International Coordinating Group (ICG) for vaccine so it can be filed the moment the epidemic threshold is crossed and the serogroup is confirmed. ICG rules do not allow filing at the alert threshold, which is why the draft has to be ready beforehand. **The local case:** in the 2024/25 season Mai'Adua had 14 suspected cases and 6 deaths, a 43% case fatality rate, against 10% for Katsina State (302 cases, 30 deaths) and 7.65% nationally. NCDC's own report lists shortages of "ceftriaxone, TI media, Lumbar Puncture kits." Nationally, 61% of confirmed cases were serogroup C and 33% W.

* Suggested formulation: check with a health worker before sharing.

Transferability: the evidence is local, which makes this the strongest action in the report. Two corrections apply. No study gives a measured cut in fatality from earlier treatment; the benefit rests on plausibility. WHO's April 2025 guidelines prefer a 5-day ceftriaxone course and allow a single dose only in large, laboratory-confirmed epidemics. The 2017 response delays (vaccination 9 weeks after the threshold, 41% of doses delivered) were measured in Zamfara, Sokoto, and Yobe, not Katsina. The state agency says drugs were already pre-positioned in May 2026, so **the first step is a stock count at each Mai'Adua clinic**, not a purchase.*

2. Ask the district and village heads to announce a standing household rule before December: a child or woman with fever and a stiff neck goes to the clinic that day, backed by a named roster of motorbike and keke drivers

PARTIALLY CONFIRMED | FRONTLINE · EMERGENCY | ADAPTATION INVENTORY: NOT CONSULTED

OWNER	BY WHEN	COST	PREVENTS
Mai'Adua district head and village heads (Daura Emirate); a volunteer driver roster	Announced by 1 Dec 2026	Near zero; fuel for drivers is unfunded	The delay while a husband is away at the Sunday market

Most married Hausa women leave home mainly for medical care, with their husband's permission. The frontline lens's point is that the dangerous delay is not the distance to the clinic but waiting for permission, so the permission is given in advance, publicly, by the people households listen to. The driver roster follows the Emergency Transport Scheme model: in Adamawa, 2013–18, 741 volunteer drivers took 18,873 women to facilities, and 99% of trips arrived within an hour.

* Suggested formulation: check with a health worker before sharing.

Transferability: the seclusion evidence is an older ethnography from Kano, not Katsina data, and no permission-barrier figure for Katsina was found. The transport evidence comes from obstetric emergencies with no control group, and drivers there were reluctant to carry older children, so the roster must name child fever explicitly.

3. Before ordering anything, ask UNICEF Katsina whether Mai'Adua is inside the therapeutic-food pipeline and the Child Nutrition Fund; then stock its outpatient sites before June 2027

PARTIALLY CONFIRMED | PUBLIC HEALTH · EMERGENCY · HISTORIAN · SKEPTIC · PHILANTHROPIST | ADAPTATION INVENTORY: NOT CONSULTED

OWNER	BY WHEN	COST	PREVENTS
Katsina SPHCDA; UNICEF Katsina for procurement; Daura stabilisation centre for complicated cases	Question answered by Dec 2026; order placed by Mar 2027; stocked by May 2027	~$136,000 for 1,700 children (see funding)	Lean-season child deaths when therapeutic food runs out

In 2024, ready-to-use therapeutic food (RUTF) ran out in Katsina, Sokoto, and Zamfara from June to October, the peak of the lean season, because of a funding gap, and that year the Katsina pipeline covered only 15 violence-affected LGAs where in 2023 it covered the whole state. MSF recorded more than 800 child deaths in its Katsina facilities from January to September 2024. In the stabilisation centres at Daura, Baure, Dutsi, and Zango, which border Mai'Adua, 42% of admissions came in May–August, and mortality was 4% (6,925 children, 2010–16). The 2025 SMART survey puts severe acute malnutrition at 2.0% (weight-for-height) to 3.4% (combined) in the northern Katsina domain that probably includes Mai'Adua. The Integrated Food Security Phase Classification (IPC) rates Mai'Adua Phase 3, Serious. Local procurement has cut the lead time from 6 months to 3, so a March order is workable.

* Suggested formulation: check with a health worker before sharing.

Transferability: the stock-out pattern and the treatment outcomes are from this state and its neighbours. Two caveats. The cost-per-life figure the funder lens first used ($1,760, Lusaka 2005–08) is a model comparing treatment with no treatment, so it is weak evidence for rural Katsina today. And the state reports that outpatient sites already exist in Mai'Adua; whether they are stocked is the open question.

4. Female volunteers visit compounds to teach meningitis danger signs, oral rehydration, and arm-band (MUAC) malnutrition screening; mallams agree to send any Almajiri boy with fever to the clinic the same day

UNVERIFIED | FRONTLINE · PHILANTHROPIST · HISTORIAN | ADAPTATION INVENTORY: NOT CONSULTED

OWNER	BY WHEN	COST	PREVENTS
Save the Children CORE Group Partners Project (CGPP), if still funded; otherwise ward health committees	From Nov 2026, through the season	~$50,000, an unsourced assumption	Cases that official channels never see

Women volunteers can enter compounds where men cannot, and the meningitis season (roughly October to April) hits ages 5–14 hardest, which makes Qur'anic school boys, sleeping crowded and away from their parents, a sensible target. Save the Children's CGPP names Mai'Adua among its LGAs and announced mass MUAC screening there, and its polio volunteers are women who work house to house.

Why unverified: CGPP was USAID-funded and most Nigerian USAID awards ended in 2025; no evidence was found that it is still running. No effect size was found for the mallam-escalation step. Screening without stocked therapeutic food (action 3) only produces referrals nobody can treat.

5. One health line in each Friday sermon, and a volunteer table at the Sunday market that starts a local missed-vaccine list

PARTIALLY CONFIRMED | FRONTLINE · SKEPTIC | ADAPTATION INVENTORY: NOT CONSULTED

OWNER	BY WHEN	COST	PREVENTS
Imams through UNICEF's Faith for Life network; ward volunteers	From Nov 2026	Near zero	Missed vaccination and late care-seeking

* Suggested formulation: check with a health worker before sharing.

UNICEF's Faith for Life programme already works through religious leaders in all 34 Katsina LGAs. The state's tracking of children who miss vaccines (Majalisa Rigakafi) covers only 8 LGAs, and Mai'Adua is not one of them, so a homemade list adds something real. Katsina's routine immunisation is weak: third-dose pentavalent coverage was 53.3% (NDHS 2024).

Corrections: the 2003–04 polio boycott was in Kano, Zamfara, and Kaduna, not Katsina, and it ended through dialogue and independent vaccine testing more than through traditional leaders alone. The market table depends on cross-border trade still running: Mai'Adua's livestock market relies on Niger, and the February 2026 border reopening covered only two Kebbi State crossings.

6. Repair Mai'Adua's flood-damaged water points and place oral rehydration salts, zinc, and cholera kits at its clinics during the dry season

PARTIALLY CONFIRMED | EMERGENCY · PUBLIC HEALTH · HISTORIAN | ADAPTATION INVENTORY: NOT CONSULTED

OWNER	BY WHEN	COST	PREVENTS
Katsina RUWASSA (water points); Katsina SEMA; SPHCDA (kits)	Before the 2027 rains (by May)	Kits ~$20,000, an unsourced estimate; repairs unfunded	Cholera deaths after the next floods

The September 2024 floods affected 123,377 people across 17 Katsina LGAs, including the Mai'Adua wards 'A' (536 people), 'B' (689), and 'C' (686), as well as Koza (847) and Natsalle (297); 127 water points were destroyed statewide. In the 2021 outbreak Katsina recorded 5,796 cholera cases and 182 deaths (fatality 3.1%), with over-60s worst affected (6.9%).

* Suggested formulation: check with a health worker before sharing.

Corrections: Bumbum does not appear in the flood record, the 127 water points are statewide, and the 2021 figures come from a conference abstract. Cholera is a rainy-season disease, so this action works after the El Niño peak, not during it.

7. Prepare heat messaging for March–May 2027 on Hausa radio, in mosques, and by volunteers: wet the skin and clothing, keep stored water, rest at midday

CONFIRMED · SKEPTIC · PUBLIC HEALTH · ADAPTATION INVENTORY: NOT CONSULTED

OWNER	BY WHEN	COST	PREVENTS
Katsina SEMA with SPHCDA; NiMet for forecasts	Messages ready by Feb 2027	Low; not costed	Heat illness in the real hot season

Katsina's hot season is March to May: mean daily maxima of 36.8°C in March and 39.3°C in April, with humidity of 14–23%. December–February, the El Niño peak, is the cool harmattan (29.5°C in January). Skin wetting is the cooling advice that works in very dry heat without electricity.*

Transferability: the temperatures are Katsina city's NOAA normals, about 100 km from Mai'Adua. No study measures the effect of heat messaging in this setting.

8. Check that Mai'Adua's poorest households are on the national cash-transfer register before the lean season

PARTIALLY CONFIRMED · ECONOMIST · PHILANTHROPIST · ADAPTATION INVENTORY: NOT CONSULTED

OWNER	BY WHEN	COST	PREVENTS
Katsina State cash-transfer unit; LGA social welfare officers	By Mar 2027	Paid by existing programmes	Families cutting food before the 2027 harvest

The World Bank's $800 million National Social Safety-Nets Project scale-up pays three tranches of ₦25,000. It had reached 9.18 million households by 31 March 2026, with about $55 million undrawn. A separate conditional cash transfer started in Katsina on 16 September 2026 (3,000 households, Kaita LGA). This replaces the funder lens's proposal for a new $240,000 cash programme, which could not be verified.

* Suggested formulation: check with a health worker before sharing.

Unconfirmed: whether Mai'Adua households are on the register, and when payments land. ₦25,000 is a small transfer; this is a top-up, not protection.

9. Ask Welthungerhilfe and Oxfam to renew the lapsed Baure drought plan and add Mai'Adua, with a fixed rainy-season trigger

PARTIALLY CONFIRMED | ECONOMIST | ADAPTATION INVENTORY: NOT CONSULTED

OWNER	BY WHEN	COST	PREVENTS
Welthungerhilfe (WAHAFA facility, German Federal Foreign Office funds), Oxfam Germany, CRUDAN	Before NiMet's Feb 2027 seasonal forecast	Baure plan: €130,447 for 326 households	A failed 2027 harvest turning into hunger

The only forecast-triggered money found anywhere near Bumbum is a drought plan for neighbouring Baure LGA (five communities, 1,956 people) whose validity ran out in April 2026. Phase 1 triggers on NiMet's February seasonal forecast of 250–450 mm. Most of the money is farm inputs (€64,800) and staff (€35,000); only €21,175 is cash.

Corrections: the Phase 2 trigger needs a NiMet forecast of at least 45°C for 7 days during the rains, which has no recorded precedent at Katsina, so it would probably never fire; it needs rewriting. Whether the facility's 2026–29 successor exists, or includes Nigeria, is unverified. The case for this plan is the area's general drought risk, not El Niño: the El Niño link to Sahel rainfall is inconsistent, and this event will be fading by the 2027 rains.

10. Ask the UN Humanitarian Coordinator to put Nigeria's north-west forward for a CERF proactive allocation, triggered by a March 2027 Cadre Harmonisé projection of Phase 3+ for Katsina

UNVERIFIED | ECONOMIST | ADAPTATION INVENTORY: NOT CONSULTED

OWNER	BY WHEN	COST	PREVENTS
UN OCHA Nigeria; advocacy by partners	Request before Mar 2027	Not set by any source	Money arriving after the lean season instead of before it

* Suggested formulation: check with a health worker before sharing.

OCHA's June 2026 El Niño portfolio lists drought frameworks for Niger ($8 million, being updated), Chad, Burkina Faso, and Mauritania, and says proactive allocations are "being prepared" for high-risk countries without frameworks. Nigeria is not mentioned. Nigeria's only CERF framework is $7 million for floods in 7 Adamawa LGAs. The Nigerian Red Cross early-action protocol covers floods in 11 states, not Katsina. Nigeria holds no African Risk Capacity policy.

Why unverified: nothing found shows Nigeria is a candidate. This is advocacy with a stated trigger, not money that exists.

Lower priority: enrolling vulnerable households in the Basic Health Care Provision Fund

PARTIALLY CONFIRMED ECONOMIST

Katsina's scheme has 583,460 enrollees, 184,105 of them funded by the Basic Health Care Provision Fund (BHCPF), with a ₦243 million state counterpart paid. Enrolment is worth doing for malaria, antenatal care, and deliveries, but it adds nothing for the two biggest killers here: meningitis outbreak treatment and severe malnutrition care are already free at government facilities.

* Suggested formulation: check with a health worker before sharing.

WHO DOES WHAT

RACI CHART

R = Responsible (does the work) · **A** = Accountable (one per action; signs off and answers for it) · **C** = Consulted (asked before) · **I** = Informed (told after). Built from the owners named in the verified actions. No one here has been contacted or has agreed to a role; every row is a proposal.

Action	Katsina SPHCDA	Mai'Adua LGA (PHC, DSNO, social welfare)	NCDC	Katsina SEMA	District & village heads	Imams	Women volunteers (CGPP or ward committees)	Driver roster	UNICEF Katsina	RUWASSA	State cash-transfer unit	Welthungerhilfe / Oxfam / CRUDAN	OCHA Nigeria	NiMet
1. Meningitis stock, vehicle, reporting	**A**	**R**	**R**	**R**	**I**	**I**	**I**	**C**						
2. Same-day care rule and drivers	**C**	**C**			**A/R**	**R**	**I**	**R**						
3. Therapeutic food pipeline	**A**	**R**			**I**		**C**		**R**					
4. Compound visits and mallams	**C**	**C**			**C**	**C**	**A/R**		**C**					
5. Sermons, market table, vaccine list	**C**	**R**			**C**	**R**	**R**		**A**					
6. Water points and cholera kits	**R**	**C**	**I**	**A**	**I**					**R**				
7. Heat messaging, Mar–May	**R**	**I**		**A**	**I**	**R**	**R**							**C**
8. Cash-transfer register		**R**			**C**						**A**			
9. Renew the drought plan		**C**		**C**	**C**							**A/R**		**C**
10. CERF proactive request				**C**					**C**			**C**	**A/R**	**I**

What the chart shows. **The State Primary Health Care Development Agency is accountable for the two actions that save the most lives** (meningitis and therapeutic food), so a single meeting with it covers most of the value in this report. **The district head is the only party both accountable and doing the work** (action 2): the palace is the front door. **Nothing in the chart is funded by a trigger**: the only forecast-triggered money near Mai'Adua lapsed in April.

WHERE MONEY DOES THE MOST

FUNDING

The Philanthropist lens, reconciled against the ranked actions and verified. Placements are ranked by marginal lives saved per dollar. The envelope is bounded by what this place can spend before the peak, not by what it needs. No pre-arranged money covers Mai'Adua: Nigeria is not in OCHA's El Niño portfolio, the Red Cross protocol is for floods in other states, and the Baure drought plan lapsed in April 2026. Seasonal malaria chemoprevention is already paid for in Katsina by the Global Fund, so none is placed here.

* Suggested formulation: check with a health worker before sharing.

FLOOR	TARGET	CEILING	LARGEST UNCERTAINTY
$120,000	$216,000	$592,000	How many Mai'Adua children existing outpatient sites, the Daura stabilisation centre, and the Child Nutrition Fund already cover

Malnutrition caseload: 201,178 (2006) × 1.03^{20} ≈ 363,000 people in 2026 (3% annual growth, our assumption) × 18% under five ≈ 65,400 children × 2.0–3.4% severe acute malnutrition (2025 SMART survey, northern Katsina domain; Mai'Adua's inclusion is probable, not confirmed) × 2.6 prevalence-to-incidence factor ≈ 3,400–5,780 cases a year. The survey ran in the peak season, so the factor may overstate the annual count. Unit cost: $67.07 per child treated (2019, Sokoto trial, therapeutic food included; RUTF is over 70% of it) × ~1.2 for inflation (our assumption) ≈ $80. Floor = 1,000 children × $80 + $40,000 meningitis = $120,000. Target = 1,700 × $80 ($136,000) + $60,000 meningitis + $20,000 cholera kits = $216,000. Ceiling = 5,780 × $80 ($462,400) + $60,000 + $20,000 + $50,000 screening (only if CGPP is running) ≈ $592,000. The funder lens proposed $500,000 for the target; verification replaced its $219 per-child-cured figure (2014–15, other states) with the per-child-treated Sokoto figure, and dropped $240,000 of cash transfers that no recipient could be found for. **The target fell 57%.**

RECIPIENT	USE	AMOUNT	MECHANISM	BASIS	VERDICT
Katsina SPHCDA, with UNICEF Katsina procurement	Action 3: therapeutic food and outpatient treatment for 1,000–1,700 children	$80,000–136,000	Pre-positioned stock, through the state pipeline	$67.07 per child treated, Sokoto 2018–19 trial (PMC10022243), inflated ~20%; RUTF $45.40 a carton (UNICEF 2025)	PARTIALLY CONFIRMED
Katsina SPHCDA	Action 1: meningitis drugs, tests, transport	$40,000–60,000	Direct grant, after a stock count	Unsourced assumption. Vaccine for outbreaks is paid for by the ICG stockpile, so this funds operations only	UNVERIFIED
Katsina SPHCDA	Action 6: ORS, zinc, cholera kits	$20,000	Pre-positioned stock	Unsourced assumption	UNVERIFIED
Save the Children CGPP	Action 4: MUAC screening and referral	$50,000 (ceiling only)	Top-up to an existing scheme	No published unit cost; funding status after the 2025 USAID cuts unknown	UNVERIFIED

* Suggested formulation: check with a health worker before sharing.

RECIPIENT	USE	AMOUNT	MECHANISM	BASIS	VERDICT
~~Save the Children~~	~~Pre-lean-season cash, 2,000 households~~	~~$240,000~~	Dropped	No Save the Children cash programme found in Katsina; the national programme is live (action 8)	UNVERIFIED

The funder's one thing: the large malnutrition responders (MSF, the Red Cross's RUTF distribution, the EU's 60 new outpatient sites) are all in other LGAs. The cheapest life saved here is a child who is already near a stocked clinic but never gets there. Money is not the binding constraint; reach is. So the first dollar should buy an answer from UNICEF Katsina about coverage, and the second should go to whoever can screen door to door.

DO NOT DO THIS — CONTRAINDICATED

Advice that circulates for this hazard and would cost lives or waste the window here.

Parametric heat-income insurance triggered at 45°C

CONTRAINDICATED

The economist lens proposed SEWA-style heat cover (India: ~$3 premium, $4.80 base payout) with a trigger of 45°C on 2 consecutive days at Katsina's NiMet station. No record shows Katsina reaching 45°C: NiMet's highest is 44°C, other datasets show 42.2°C, and the hot season is March–May, outside the El Niño window. The cover would almost never pay. The station is also about 100 km from Mai'Adua, so basis risk is high, and no heat or index insurance operates in Katsina.

* Suggested formulation: check with a health worker before sharing.

Instead: if a sponsor wants to pilot heat cover, calibrate the threshold (around 41–42°C) against NiMet's station record first, and expect payouts of a few dollars. Otherwise put the premium money into action 1.

Promoting electric fans for the March–May heat *

DO NOT DO · PARTIALLY CONFIRMED

In dry heat above about 35°C, fans do not cool older adults and can add heat strain. A 2024 Lancet Planetary Health review advises against fans above 35°C, especially for people over 65. In a 2019 trial, fans were harmful on every measure at 47°C and 10% humidity. Katsina's April conditions are 39°C at 23% humidity. Only 39% of rural Nigerian households had electricity in 2018, so fan advice also assumes power that most homes lack.

Instead: wet the skin and clothing, use shade, and rest at midday. Fans are fine at night when the air is cooler.

Making text-message or app warnings the main channel

DO NOT DO · CONFIRMED

In Katsina, 33.6% of women own a mobile phone against 81.9% of men, and 26.6% of women are literate (NDHS 2018). An SMS-first warning reaches the men at the market and misses the women and children at home.

Instead: Hausa radio, Friday sermons, and women volunteers going compound to compound (actions 4, 5, and 7).

Waiting until the epidemic threshold to start organising a meningitis vaccine request

DO NOT DO · FROM THE RECORD

In 2017, vaccination in the north-west began about 9 weeks after the epidemic threshold against a 4-week target, and only 41% of requested doses arrived. Those figures are from Zamfara, Sokoto, and Yobe, not Katsina, but the mechanism is the same.

Instead: draft the ICG request at the alert threshold and file it the moment the epidemic threshold and serogroup are confirmed (action 1).

* Suggested formulation: check with a health worker before sharing.

YEAR	EVENT	TOLL	WHAT CHANGED AFTER
1912–15	Famine across northern Nigeria; known in Katsina as *Yar Dikko*	No Katsina figure found	Not documented
1972–74, 1983–84	Sahel droughts; 1982–83 was a major El Niño	No Katsina or Daura figure found	Great Green Wall shelterbelts later planned; Mai'Adua is a listed LGA
1996 (Jan–Jun)	Meningitis epidemic, among the largest recorded	Nigeria: 109,580 cases, 11,717 deaths. Katsina: 1,590 cases and 297 deaths by 29 Feb (WHO, not fetched directly)	Reactive campaigns of over 13 million doses, after the fact
2005	Niger food crisis spills across the border	MSF feeding over 600 under-fives at a Katsina border centre in July	CMAM introduced in Katsina in 2010
2010–11	Community malnutrition treatment (CMAM) scaled in Katsina	9,031 severe cases admitted Jan–Sep 2011; coverage in Daura and Zango only 44.6%	Stabilisation centres at Daura, Baure, Dutsi, and Zango
2017	Meningitis C epidemic	Nigeria: 14,518 cases, 1,166 deaths. Katsina third-worst (915 cases)	Katsina vaccinated in Apr–May 2017, late in the season
2021	Cholera outbreak, Katsina	5,796 cases, 182 deaths (3.1%)	Nothing specific found
2024 (Mar)	Reactive Men5CV campaign, the world's first, in seven states including Katsina	Not applicable	Whether it reached Mai'Adua is not on record
2024	Therapeutic food stock-out, Jun–Oct; malnutrition at emergency levels	MSF: over 800 child deaths in its Katsina facilities, Jan–Sep	Pipeline cut to 15 LGAs; Child Nutrition Fund launched in 10 unnamed LGAs

* Suggested formulation: check with a health worker before sharing.

YEAR	EVENT	TOLL	WHAT CHANGED AFTER
2024 (Sep)	Floods across 17 Katsina LGAs	123,377 affected, 12,927 displaced; Mai'Adua wards A, B, and C, Koza, and Natsalle hit	Relief for the 2025 floods was still being handed out in Feb 2026
2024/25	Meningitis season	Mai'Adua: 14 suspected cases, 6 deaths (43%). Katsina: 302 cases, 30 deaths	State says drugs were pre-positioned in May 2026

The recurrence nobody has noticed: every hot, dry winter in this belt is a meningitis winter, and every epidemic here has been answered late. In 1996 the doses came after the fact; in 2017 Katsina vaccinated in April and May. Each new vaccine cleared space for the next serogroup: A was eliminated, then C arrived, and now C and W dominate. The pathogen changes. The response lag does not, and nobody budgets for it.

WHERE THE EXPERTS DISAGREE

UNRESOLVED

- **"El Niño will kill ~45 people here through heat"** versus **"the deaths that are documented here come from meningitis and malnutrition, and the projection window is the cool season."** The national figure is Climate Impact Lab's; the Skeptic and the public health scientist hold the counter-claim, and note the model had little African mortality data. A heat–mortality study using Katsina death records would settle it; none exists.
- **Request preventive Men5CV vaccination** (Public health, Historian) versus **"the vaccine always arrives after the peak; fix treatment instead"** (Emergency). Both can be done, but only treatment readiness can be finished before December. Confirmation of whether Mai'Adua was reached in March 2024 would decide how much the vaccine half matters.
- **Build a drought trigger for the 2027 season** (Economist) versus **"the El Niño–Sahel rainfall link is weak, and 2015, a strong El Niño, was wet"** (Skeptic; confirmed by the financial verifier). The plan is defensible on general drought risk, not as an El Niño measure.
- **Fund cash before the lean season** (Philanthropist) versus **fund therapeutic food and reach first** (Public health, Skeptic). Verification found the national cash programme live, which shifts the cash question from funding to registration.

WHAT NOBODY COVERED

BLIND SPOT

Who actually works in Mai'Adua now. Every outreach action leans on a channel that may not exist: the CORE Group Partners Project was USAID-funded and its status after the 2025 cuts is unknown, MSF and the large food responders work elsewhere, and the state's own

* Suggested formulation: check with a health worker before sharing.

outpatient sites in Mai'Adua are reported but not confirmed. **Second, the border.** Mai'Adua lives off trade and movement with Niger, which has its own $8 million El Niño drought framework. If Niger's lean season goes badly, people and illness will cross into Mai'Adua through a border that is formally closed and therefore unrecorded. No lens could find data on that flow, and it could change both the caseload and where the clinics need stock. No lens found any data specific to Bumbum.

HOW TO READ THIS

METHOD

The panel is author-built. Seven simulated expert lenses were used to generate these recommendations. Where they agree, that is a strong hypothesis, not a consensus of the field. Every recommendation was then independently checked for evidence, transferability to this location, and feasibility before the peak; the verdict tags carry the result.

Mortality figures are resolved at Climate Impact Lab impact-region scale, roughly 300,000 people. They are not neighbourhood figures. The El Niño projection covers *heat only*; drought, flood, fire, smoke, and crop failure are additional to it and are carried here by the historical record rather than by a published number.

HOW THIS BRIEFING WAS PRODUCED

METHODOLOGY

- **Scope.** The location was taken as given: Bumbum, which resolves to wards Bumbum 'A' and 'B' in Mai'Adua LGA, Katsina State, Nigeria. Climate Impact Lab impact region, as the report records it: not resolved. Mortality figures exist only at that scale (impact regions average roughly 300,000 people); sub-city detail is used for hazard exposure and action design, not for death counts.
- **Exposure.** Climate Impact Lab, *451,000 at Risk* (September 2026, pre-peer-review): ECMWF SEAS5 seasonal forecasts (51-member ensemble) applied to temperature–mortality relationships across 24,378 regions, against a 1996–2025 baseline, for June 2026 – February 2027. It covers heat only. It is never added to the Impact Map's long-run projections or quoted from the map's GDP-denominated mortality. Where heat is not the dominant hazard, the historical record carries the hazard instead of a borrowed number. The Nigeria figures were re-checked against the live study page on 4 October 2026 and were unchanged.
- **Expert panel.** Seven simulated expert lenses (emergency manager, public health scientist, frontline responder, skeptic, economist, historian, and philanthropist) each researched this place independently and in parallel, from web sources, to a fixed brief.
- **Reconciliation.** Duplicate actions were merged and backing lenses recorded. Convergence was tested for shared bias (whether the evidence behind an agreed action came from a setting like this one) before it counted in an action's favour. Contradictions were kept, not resolved by

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dropping a side. Actions were ranked by expected lives saved per unit of effort before the peak, each was given a named payer and trigger or marked unfunded, and the funding placements were reconciled against the ranked list. The RACI chart was built from the owners named in the verified actions.

- **Verification.** Six separate verifier agents checked every recommendation on three tests: evidence (the primary source and its effect size), transferability (whether that evidence came from somewhere like here), and feasibility (whether it can be done here before the peak). Financial actions were also checked for availability in this jurisdiction, a named payer, a trigger threshold, and whether they already exist. The funding envelope was checked for real recipients, published unit costs, and duplication, then re-summed. Where two verifiers disagreed on the Mai'Adua meningitis figures, the NCDC report was downloaded and its table read directly.
- **Verdicts.** *Confirmed*: evidence, transferability, and feasibility all hold. *Partially confirmed*: holds after the corrections shown. *Unverified*: could not be checked to a primary source. *Contraindicated*: would do harm or waste the window here.
- **Adaptation Inventory.** Not consulted. Every verifier tried it; the page is a client-side tool that shows "0 adaptations" until a country is selected in a browser, and its download sits behind a sign-up form. No action in this report should be read as having no precedent there; none was checked.
- **Health wording.** Passages where a wording error could hurt someone (meningitis care, cooling and fans in dry heat, and skin wetting) are marked "* Suggested formulation: check with a health worker before sharing." They were checked against published sources, not by a health professional. They are suggestions, not medical advice, and anyone passing them on should confirm them with a doctor, pharmacist, nurse, or community health worker first. The marker was added on 2 November 2026.
- **Rules.** No cost is derived by dividing a published total, and no local figure is invented; where data is thin, the report says so and shows the containing geography's number with its arithmetic. Every source is listed below with its verification status.
- **Limits.** The panel and verifiers are AI research agents working from web-available sources, not human experts, and no one in this place was interviewed. No source anywhere is specific to Bumbum; data is at LGA, state, or national level. Several pages returned 403 and were used only through search summaries (marked below). The impact region was not resolved. Most action costs are unsourced estimates and are labelled as such. One verifier misread the NCDC table; the reading in this report is from the document itself.

References · Meningitis: NCDC CSM sitrep, epi-week 14 2025 (read directly) · CDC MMWR 66(49), 2017 outbreak (verified) · 2017 response delays, Zamfara/Sokoto/Yobe (verified, corrected) · 2009 Niger–Nigeria campaigns (verified, corrected) · WHO meningitis guidelines, 2025 (verified) · Men5CV introduction, 2024 (verified) · 1996 epidemic (abstract) · 2003–04 polio boycott (verified)
Malnutrition: UNICEF Nigeria SitRep, Dec 2024 (verified) · MSF SMART survey, 2024 (verified) · Daura/Baure/Dutsi/Zango stabilisation centres

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(verified, corrected) · CMAM coverage, 2011 (verified) · Sokoto CMAM cost trial (verified) · Bachmann, Zambia (verified, demoted) · R4D CMAM costs (verified, not used) · UNICEF RUTF prices, 2025 (verified) · North-west SMART survey 2025 (verified) · IPC Nigeria acute malnutrition (verified) · Child Nutrition Fund (verified) · Katsina OTP sites, Apr 2026 (verified) · Red Cross RUTF (verified) · MSF 2025 (verified) · CGPP expands to nutrition (verified, corrected) · CGPP Nigeria (verified) · Global Fund SMC, Katsina (verified)

Community: Schildkrout, seclusion (verified; Kano, older) · Transaid ETS, Adamawa (verified) · Faith for Life (verified) · Majalisa Rigakafi (verified) · Katsina immunisation (lens source) · Niger border reopening, 2026 (verified) · Mai'Adua market, 2023 (verified, dated) · NDHS 2018 (verified)

Cholera, floods, heat: IOM DTM, Katsina floods 2024 (read directly, corrected) · Katsina cholera 2021 (abstract) · Katsina climate normals (NOAA) (verified) · Gagnon, JAMA 2016 · Morris, Ann Intern Med 2019 · Meade, Lancet Planet Health 2024 · O'Connor, JAMA 2024 (all verified) · Nigeria electricity access (verified)

Finance: Baure drought AAP (verified, corrected) · WAHAFA evaluation (verified) · OCHA El Niño portfolio, Jun 2026 (verified) · Nigeria flood framework (verified) · Nigerian Red Cross flood EAP (verified) · ARC country engagement (verified) · SEWA heat insurance (verified) · NIRSAL–NAIC–Pula (verified) · Katsina health insurance (verified) · NASSP scale-up (verified) · Katsina CCT, Sep 2026 (verified)

History: Tarhule 1997, northern Nigeria droughts (lens source) · El Niño and Sahel rainfall (lens source) · MSF 2005 Katsina feeding centre and WHO 1996 Katsina figures (search summaries only; pages returned 403)

Location: Bumbum, Mai'Adua · Mai'Adua LGA

Exposure data · Climate Impact Lab, 451,000 at Risk · Impact Map · Adaptation Inventory

Generated 4 October 2026 for Bumbum, Mai'Adua, Katsina State, Nigeria. El Niño Ready is volunteer-run and is not affiliated with Climate Week NYC, The Climate Group, or Columbia University.

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